

The Secret of Codey Rocky



Learning Objectives

1. Understand the definition of Program and what Program can do.
2. Get to know Codey Rocky and its features.
3. Master the basics of mBlock 5.
4. Learn how to upload programs.

>>> Situated Learning

1. Have you ever seen a robot in daily life? What do you think robots are?

2. Try guessing what these robots are used for. Match the picture with the robot.



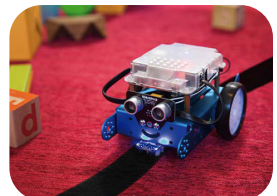
Programmable robot



Robots in logistics



Delivery robot



Security robot

3.Unveil the mysteries of Codey Rocky.

Codey Rocky is a programmable educational robot. Tiny but mighty. It can not only avoid obstacles but can also follow black lines. Using mBlock 5, you can even add more abilities to Codey Rocky, such as face recognition.



But how do we have Codey Rocky and other robots follow our instructions? Do you have any ideas? Write down your idea.



>>> Key Concept



Program

```
when Codey starts up
forever
  play sound hello
  move forward at power 50 % for 1 secs
  turn left at power 50 % for 1 secs
```

Program is an artificial language that we use to tell robots what to do. We translate our instructions into a piece of program. Then we upload the program to the robot, making it do a variety of things as programmed.



>>> Extension

Codey Rocky is an educational programmable robot. You can use software to code the robot, manipulating it to do a variety of things you can imagine. It's also a good companion that can help children learn to code. With mBlock 5, children are able to master the basics of coding and develop logical thinking as well as computational thinking. In addition, Codey Rocky supports technologies like AI and IoT, which exposes children to the latest cutting-edge technologies.



Combine Codey with Rocky and then you get a Codey Rocky. Now let's take a look at them one by one.

(1) Codey: As the brain of the robot, Codey is equipped with a variety of sensors and programmable blocks. It can work individually and can also work with Rocky to perform more tasks. Now pick up your Codey. Let's take a look at what sensors it has.



(2) Rocky serves as the chassis of Codey. It adds more abilities to Codey, like avoiding obstacles, identifying colors, following lines and more.



>>> Coding Practice

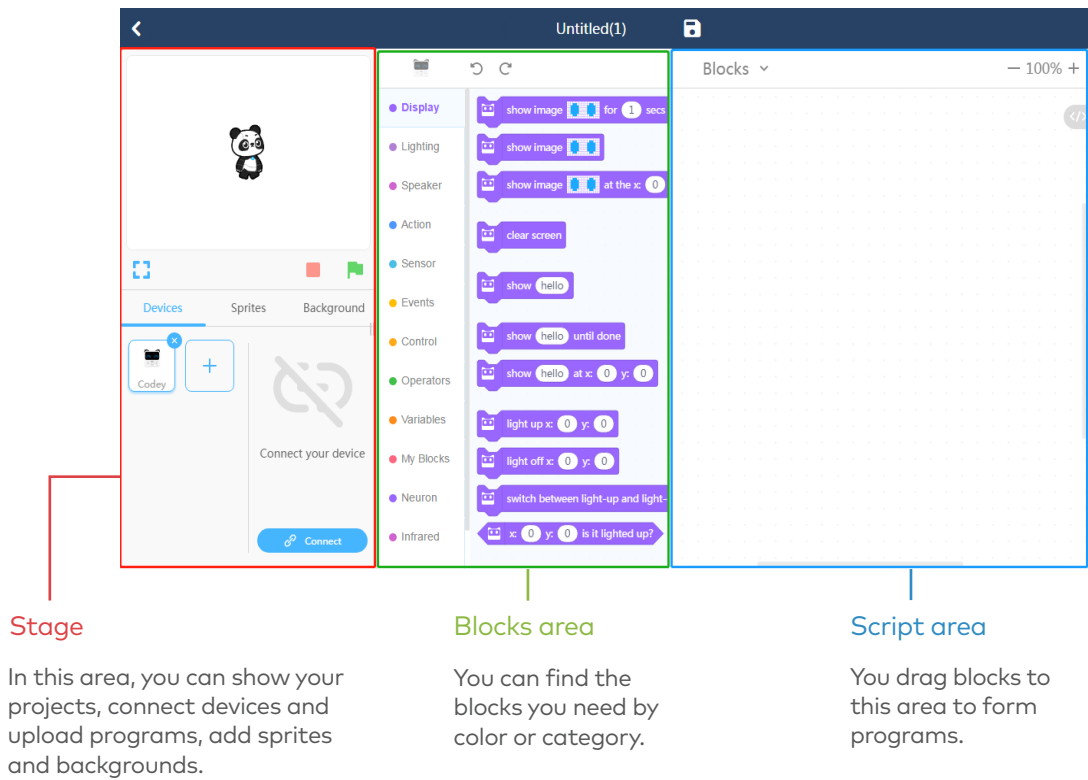
mBlock 5 is a programming tool which supports block-based and Python programming languages. It's developed based on the Scratch 3.0, an open-source software tool that's developed jointly by MIT and Google.

Using mBlock 5, you can write programs that tell Codey Rocky or other robots do whatever you would like. You can even take advantage of the software to create stories, games, and animations that are engaging and unique. Moreover, mBlock 5 exposes children to technologies, like AI, deep learning and model training. In a word, mBlock 5 can be one of the best options for first-time coders.



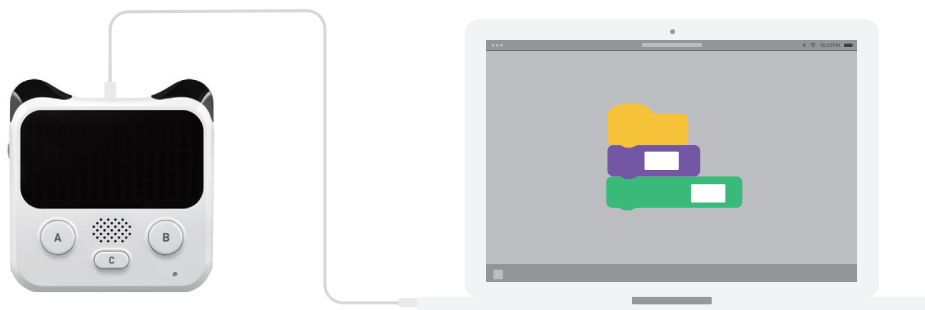
mBlock 5 Logo

Take a Tour of the Interface:

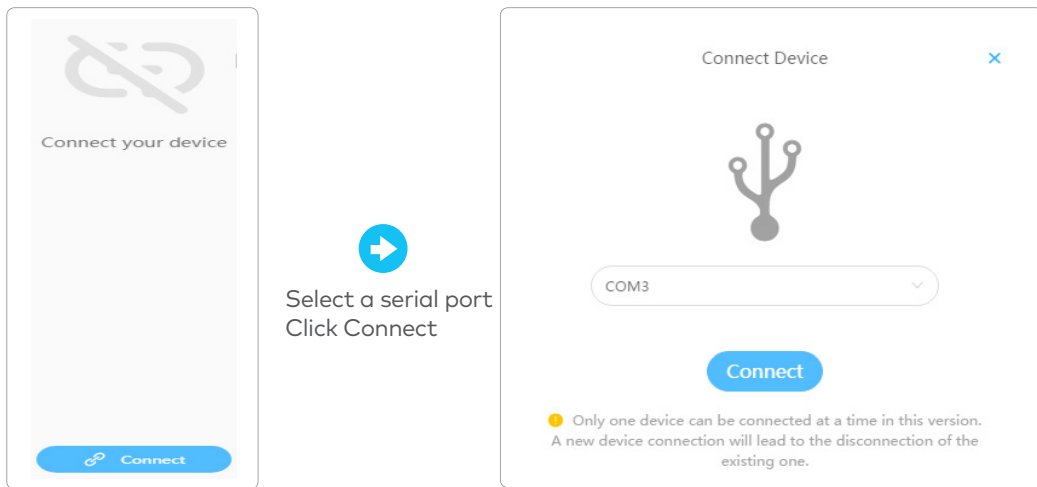


Bring Your Codey Rocky to Life

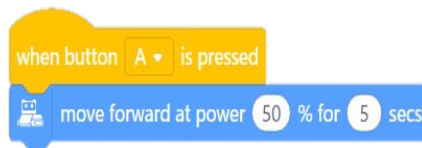
(1) Connecting to a computer: Connect Codey to the computer via the USB cable. Then power on Codey.



(2) Selecting the serial port: Open mBlock 5, click Connect, and select the correct serial port.



(3) Practice yourself! Use mBlock 5 to create the program as shown below.



(4) Upload the completed program to Codey.

(5) Unplug the data cable and place Codey Rocky on a table or the ground. Press the button A and watch how Codey Rocky reacts.

>>> Share

In this lesson, we created a project_____.

During the process, we came into an issue_____.

In the end, we solved the problem by _____.

>>> Self-review



1.In this lesson, I learned_____

2.The part I like most about this lesson is_____

3.Can you think of any situations where programs are
used in daily life?

